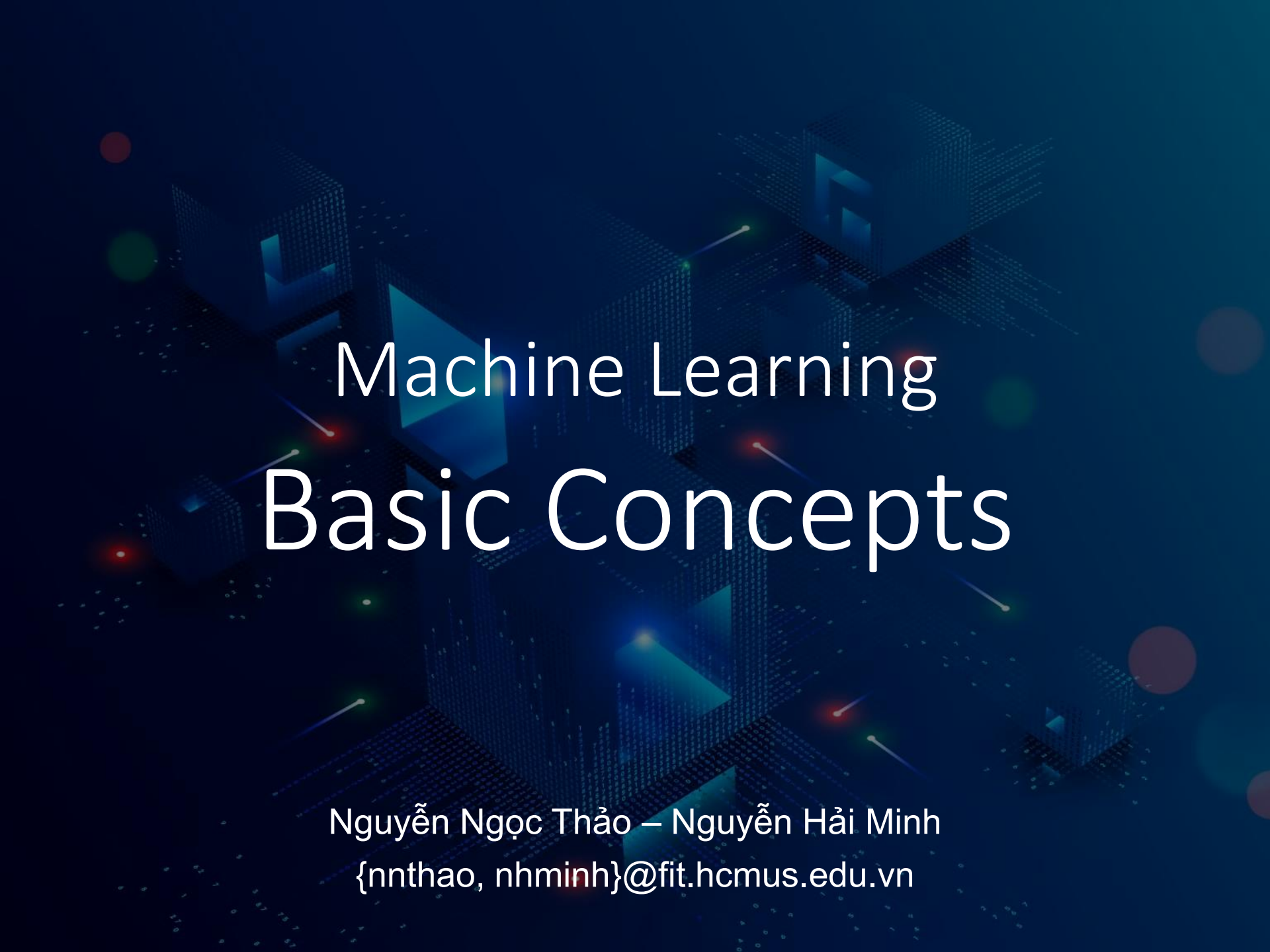


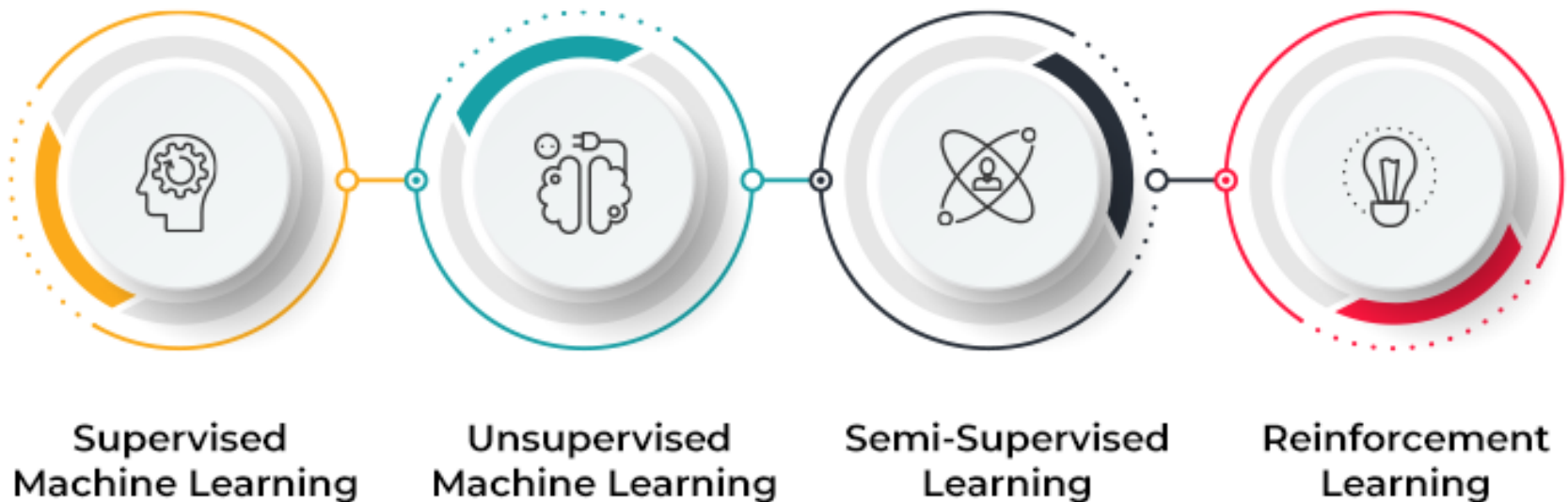


Machine Learning Basic Concepts

Nguyễn Ngọc Thảo – Nguyễn Hải Minh
{nnthao, nhminh}@fit.hcmus.edu.vn

What is machine learning?

- **Machine learning** provides machines the **ability to learn** from past experiences to **identify patterns and make predictions**.
- ML approaches are traditionally categorized into



Why learning agents?

The designers cannot anticipate all possible future situations

- A robot for maze navigation must learn the layout of each new maze it encounters.

Environment changes over time

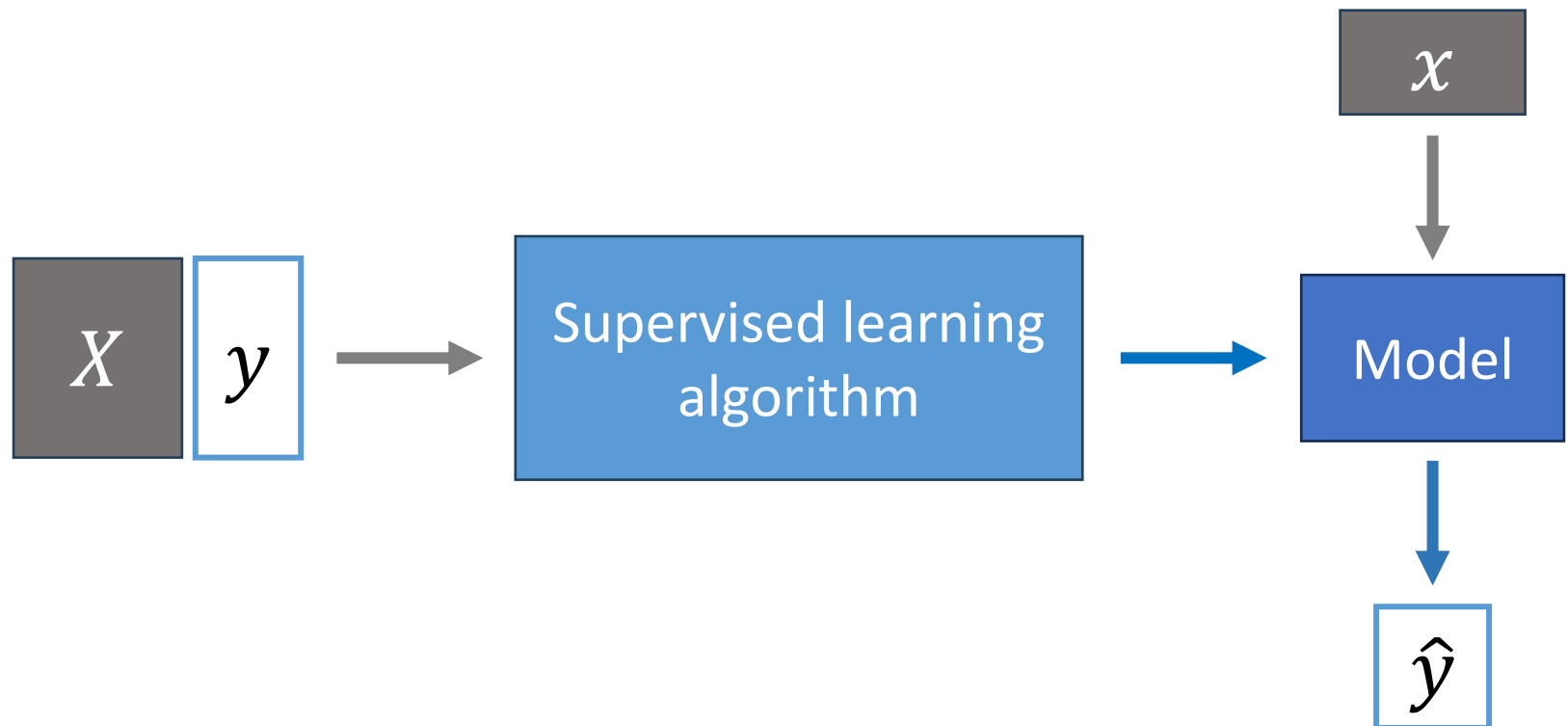
- An agent for stock market prices prediction must be adaptive to sudden changes, from boom to bust.

No idea how to program a solution

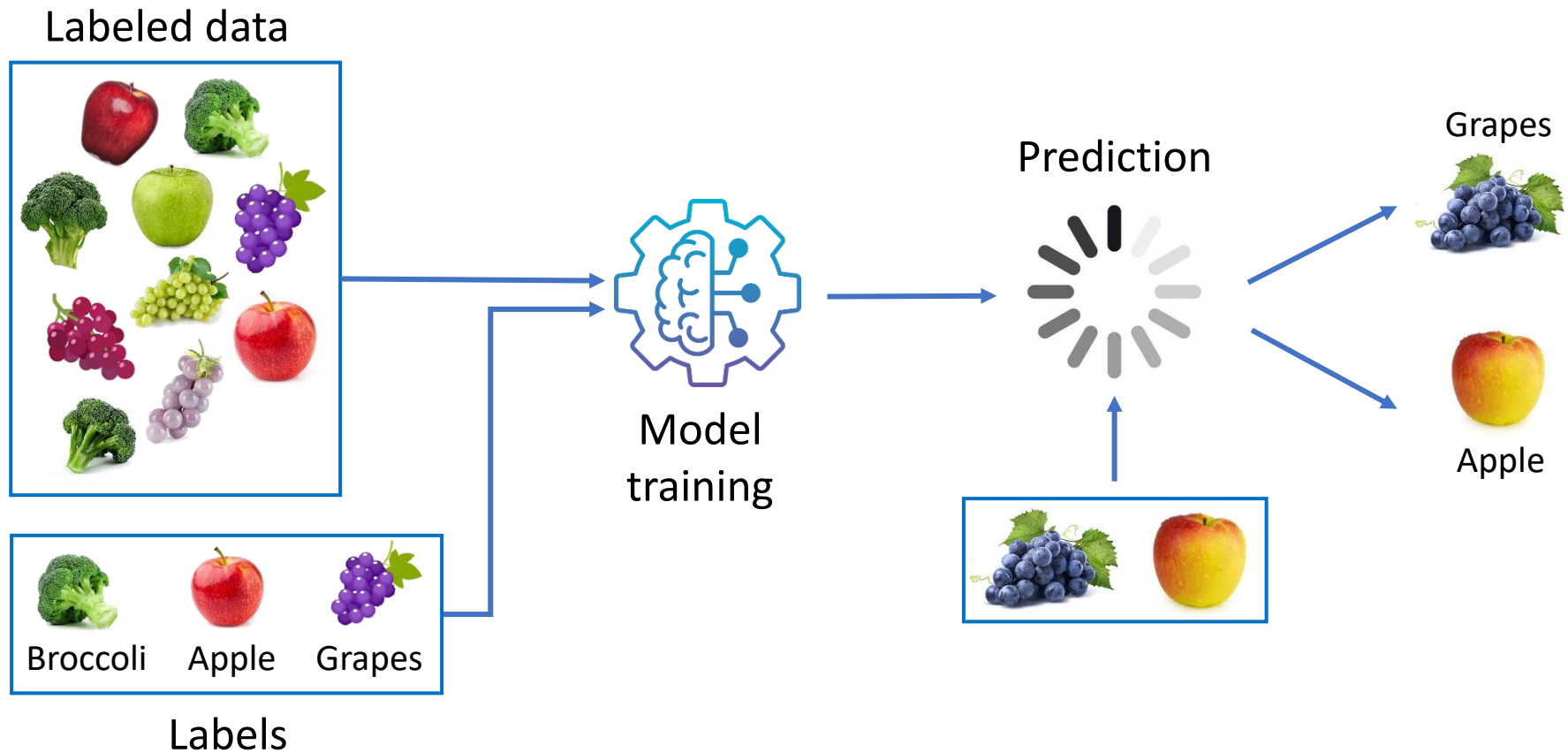
- Face recognition, fingerprint recognition, etc.

Supervised learning (SL)

- A **SL model** learns from labeled examples to map from inputs to outputs.
- Each example is a pair of input and output values.



Supervised learning (SL)



Classification vs. Regression

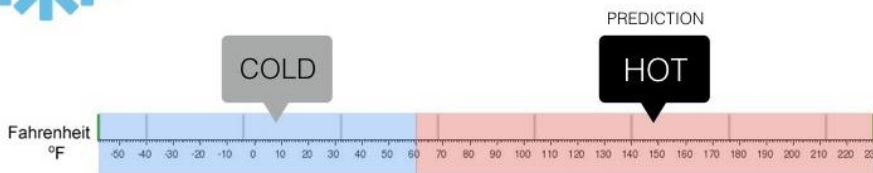
Regression

- Train a model to predict a continuous dependent variable



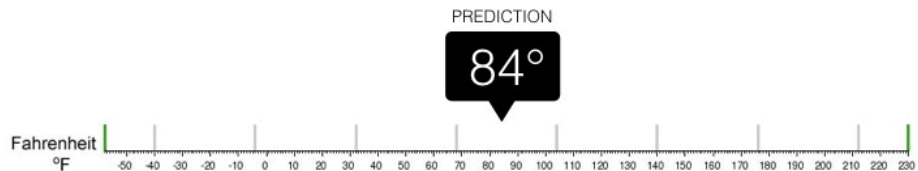
Classification

Will it be Cold or Hot tomorrow?



Regression

What is the temperature going to be tomorrow?

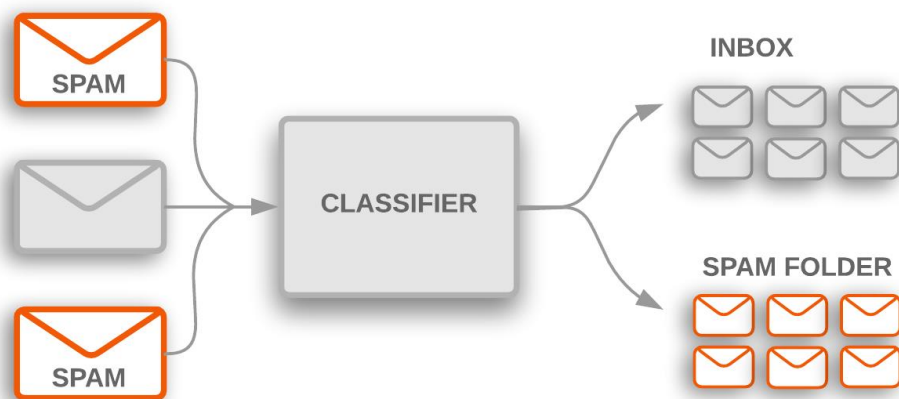


Classification

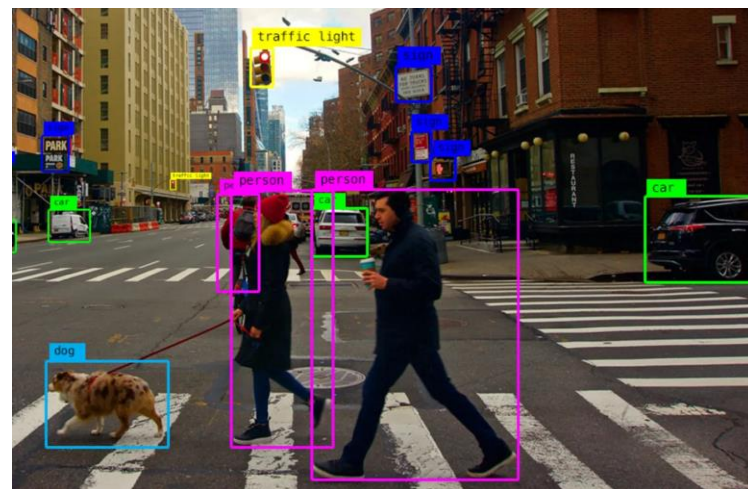
- Train a model to predict a categorical dependent variable

Classification: Problem types

- A **binary classification** task involves only two classes, while a **multi-class classification** one handles more than two classes.



Spam detection



Object detection

Classification: Problem types

- **Single-label classification:** Each example is assigned to only one class from a set of mutually exclusive classes.
 - E.g., a fruit is either an "Apple" or "Banana," but not both.
- **Multi-label classification:** Each example can be at once assigned to multiple classes that are not mutually exclusive.
 - E.g., an image tagged as "Beach" and "Sunset."



Cardiomegaly, Effusion



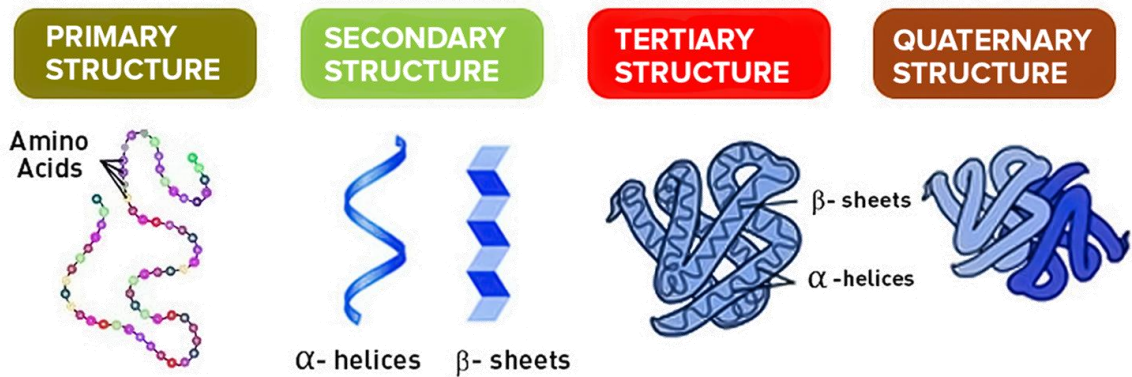
Edema



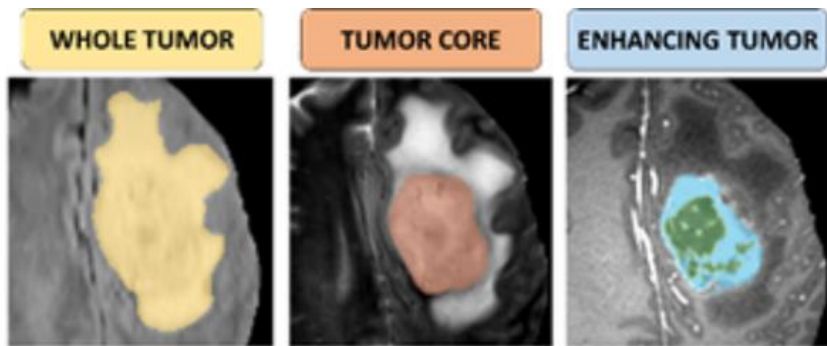
Cardiomegaly, Effusion,
Edema



No Finding

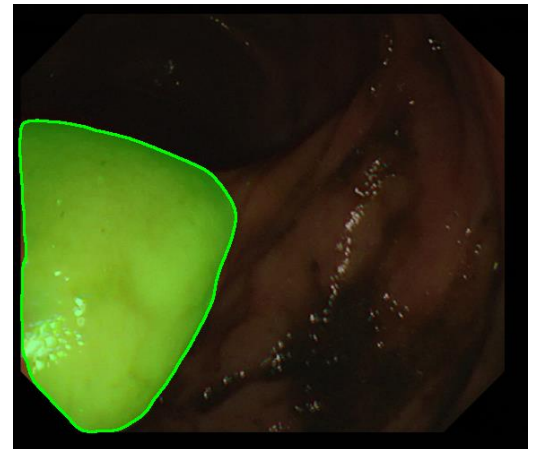
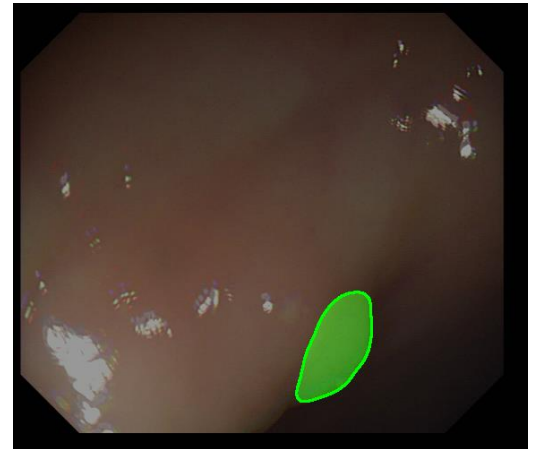
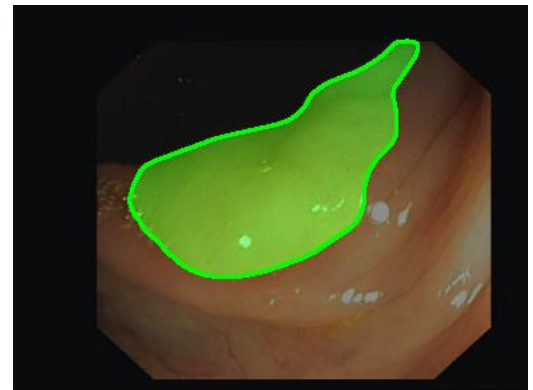


Classify protein structures at different levels



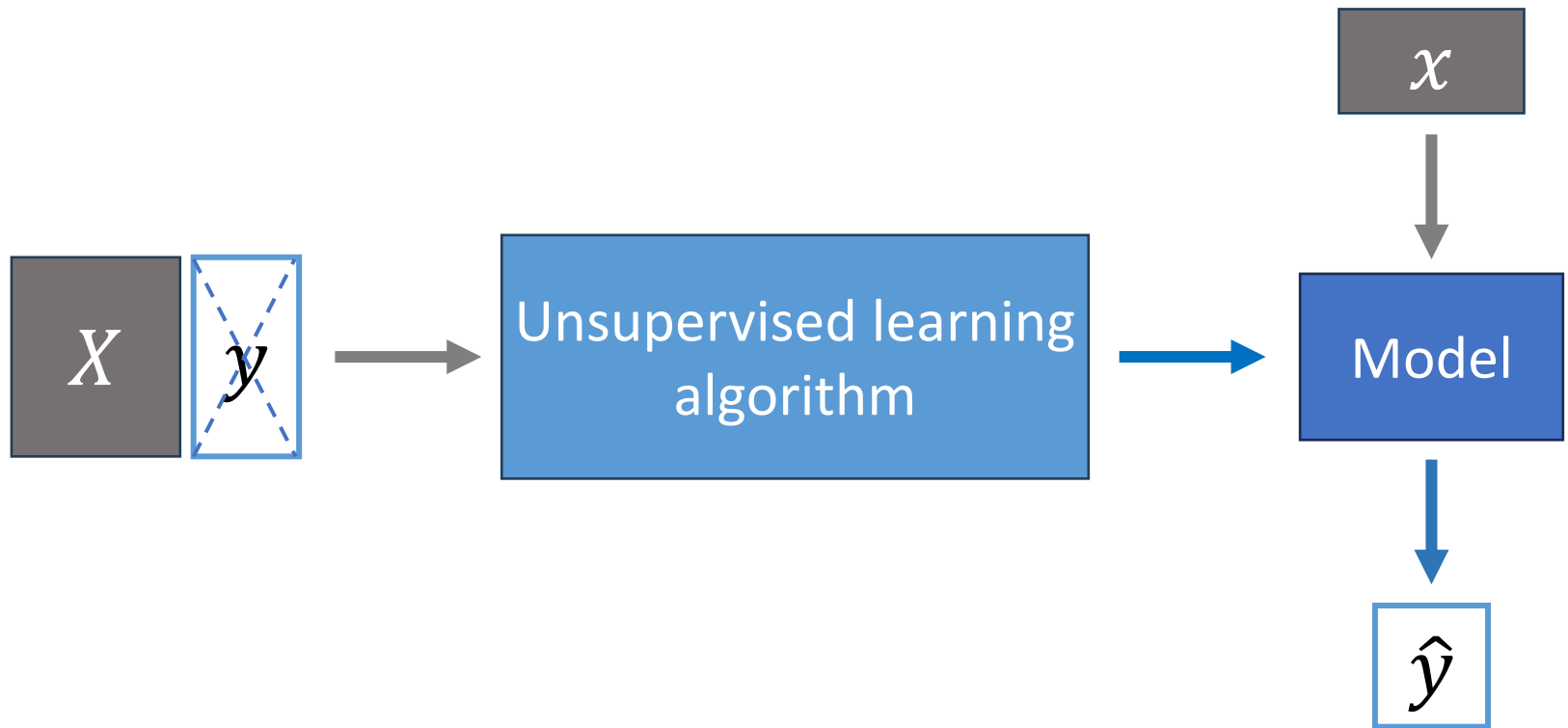
Tuan-Luc Huynh, Thanh-Danh Le, Van-Tam Nguyen, Trung-Nghia Le, and Minh-Triet Tran. "Efficient 3D Brain Tumor Segmentation with Axial-Coronal-Sagittal Embedding". PSIVT

Bui, Nhat-Tan, Dinh-Hieu Hoang, Quang-Thuc Nguyen, Minh-Triet Tran, and Ngan Le. "MEGANet: Multi-Scale Edge-Guided Attention Network for Weak Boundary Polyp Segmentation." WACV 2024. (Arkansas Uni. - HCMUS)

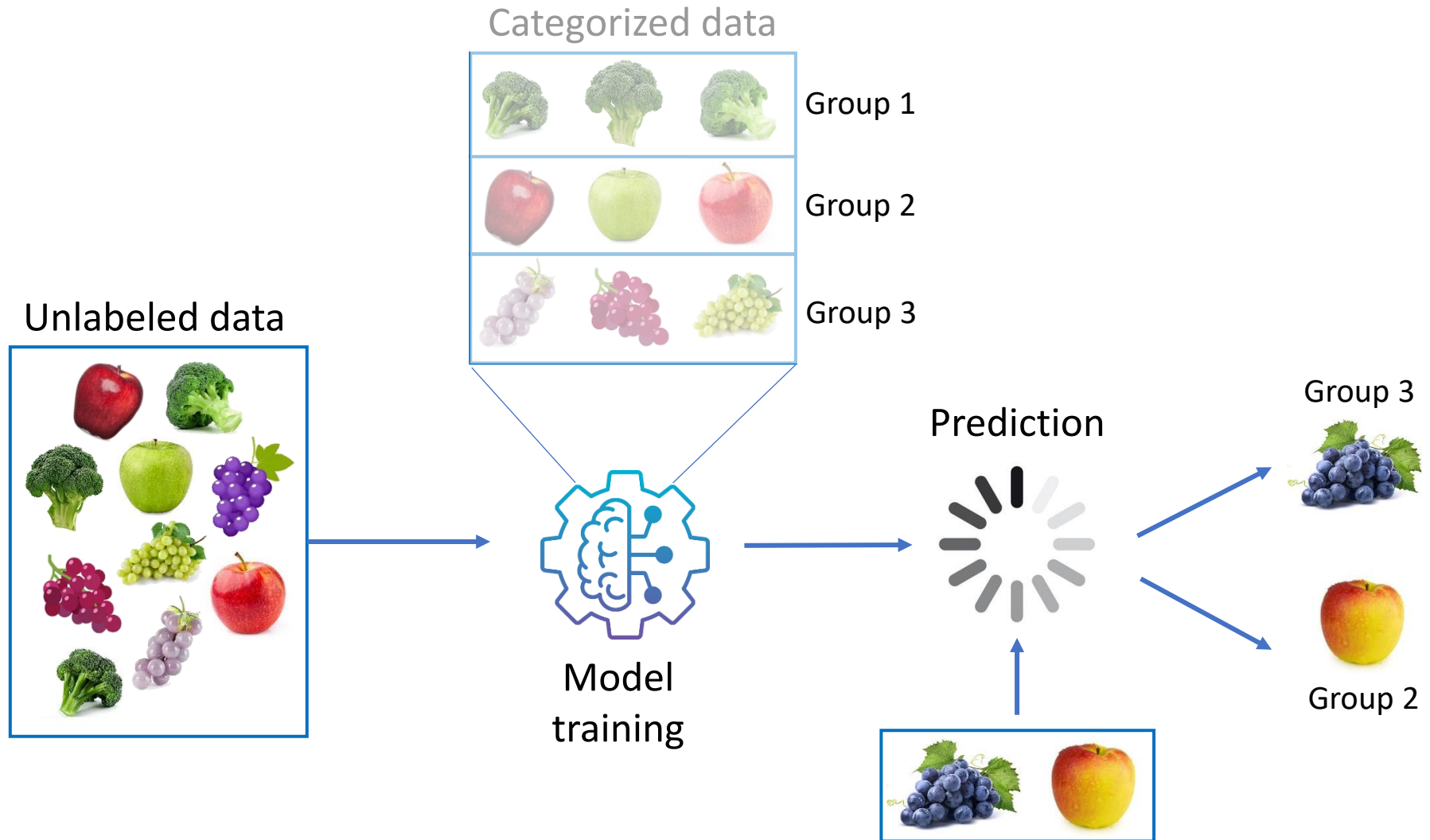


Unsupervised learning (USL)

- A **USL model** learns from unlabeled examples to describe patterns and insights in the data.

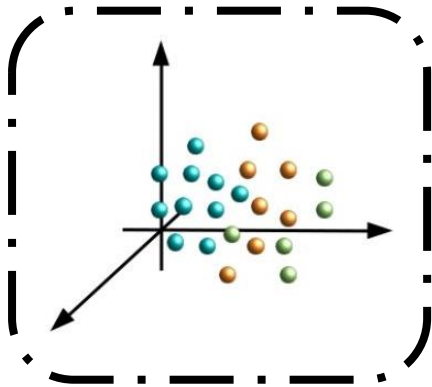


Unsupervised learning (USL)



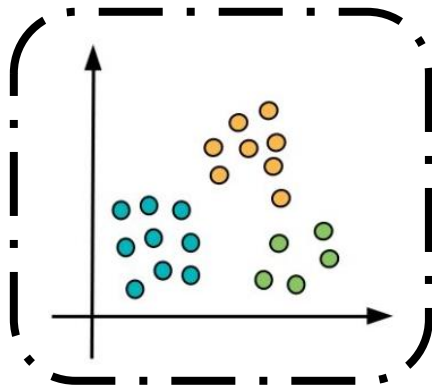
Common unsupervised learning tasks

High dimensional data



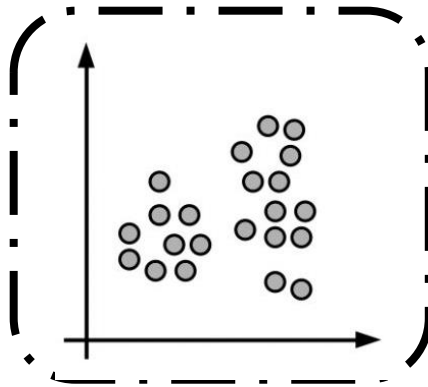
Dimensionality

Reduction

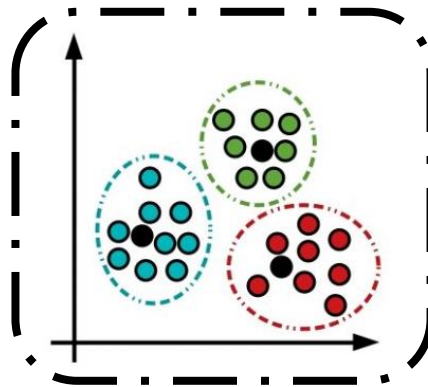


Low dimensional data

Raw data points



Clustering



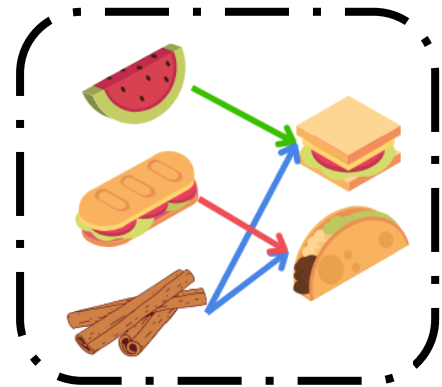
Clustered data points

Transactional data

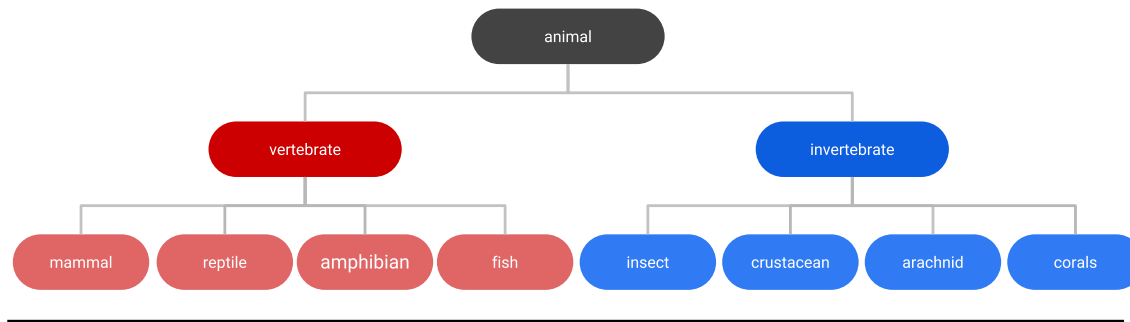


Association

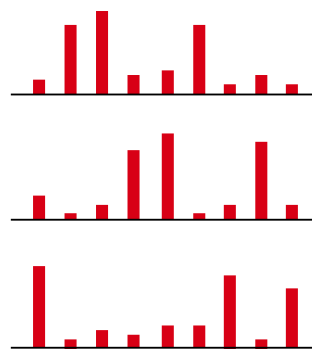
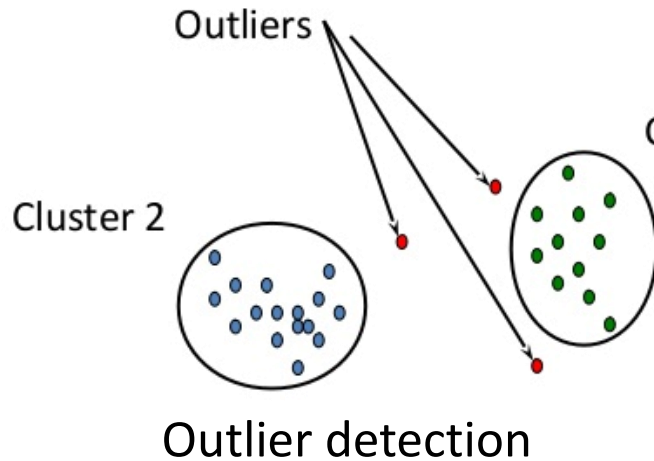
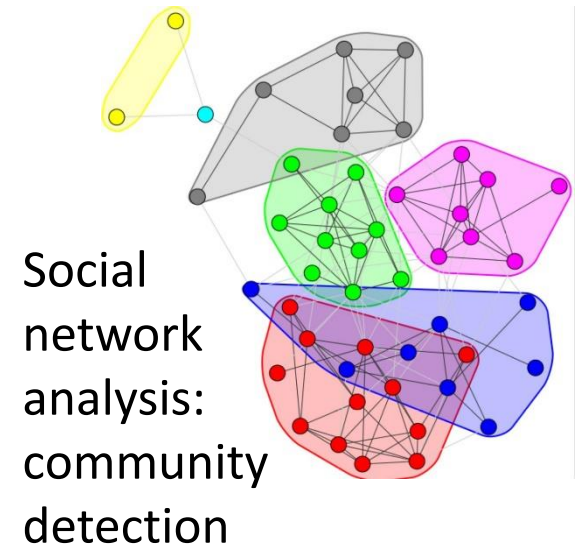
Analysis



Causal relationships



Biology: a taxonomy of living things



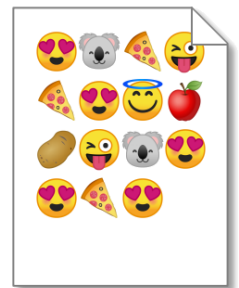
14 % Topic 1



33 % Topic 2

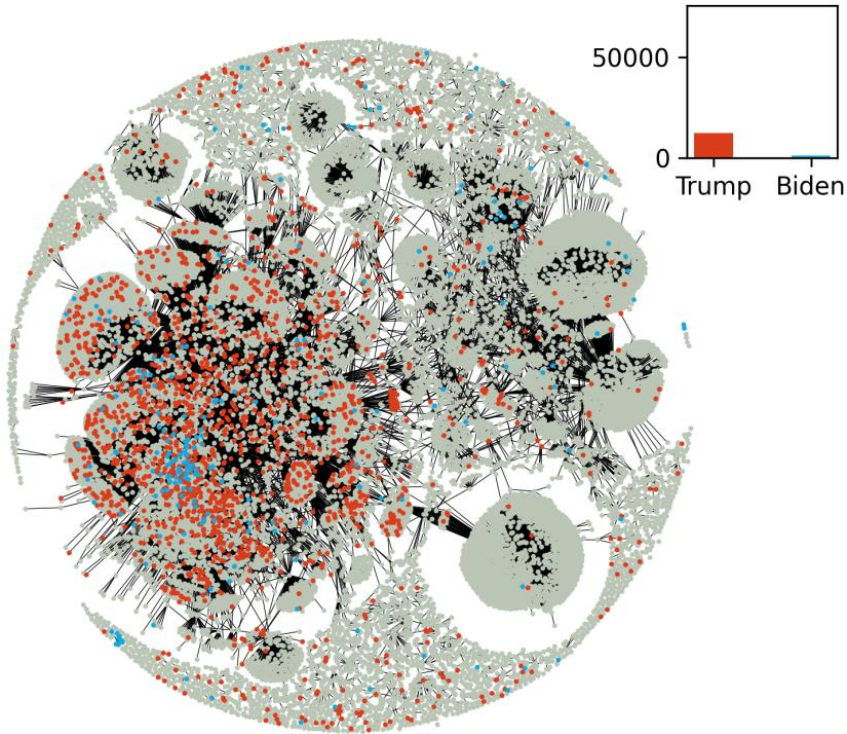


53 % Topic 3

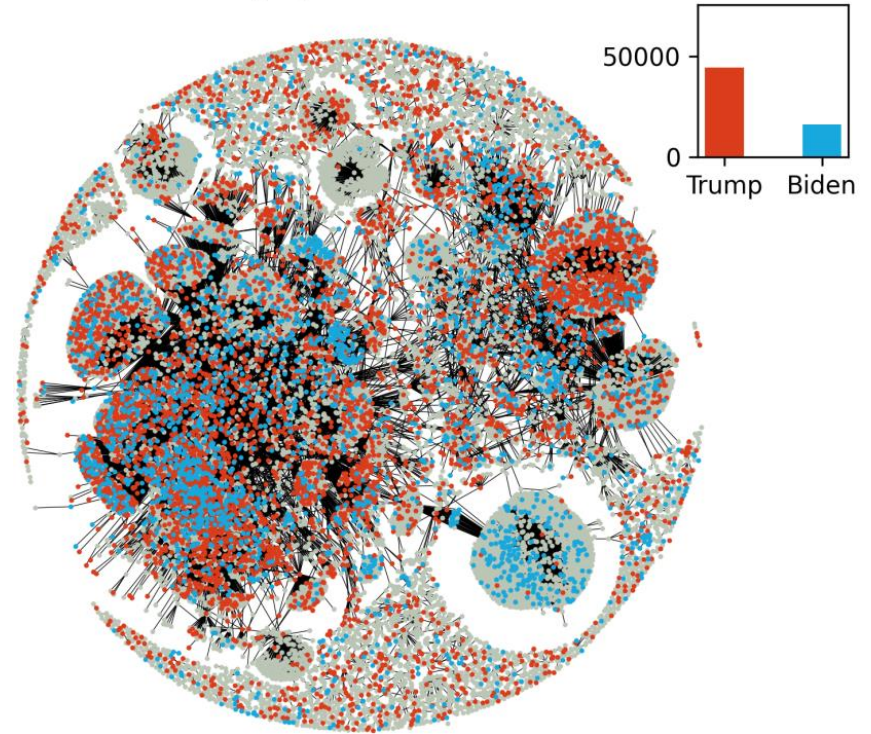


Information retrieval: document clustering

Retweet graph date: 19/07/2020



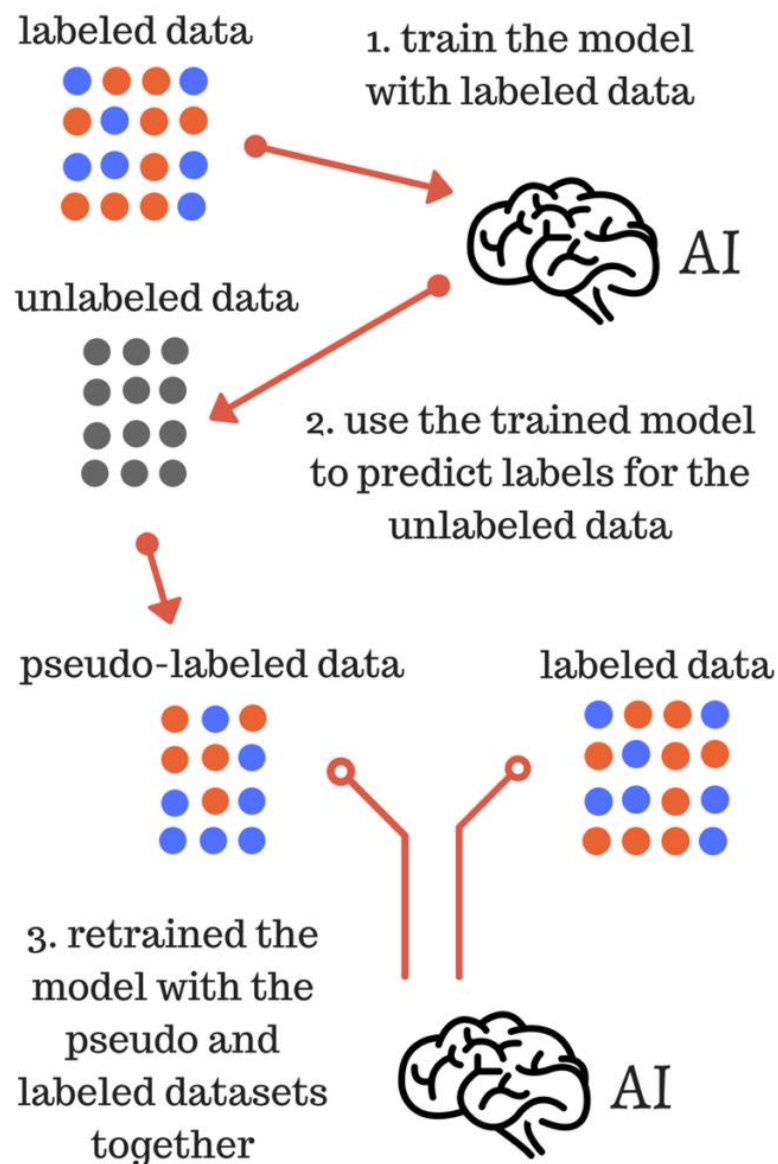
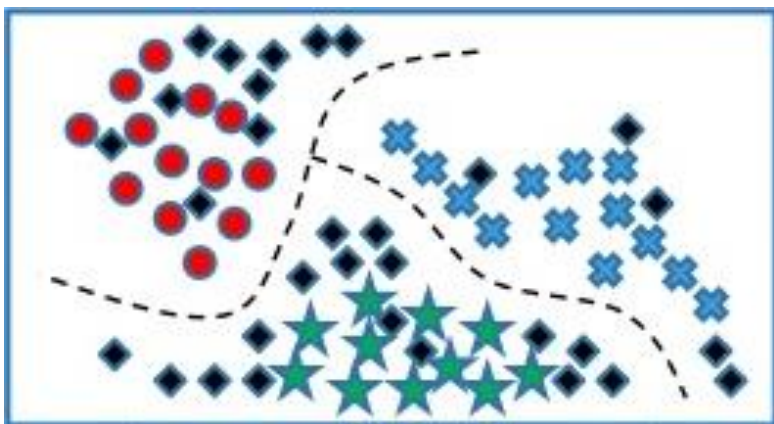
Retweet graph date: 01/09/2020

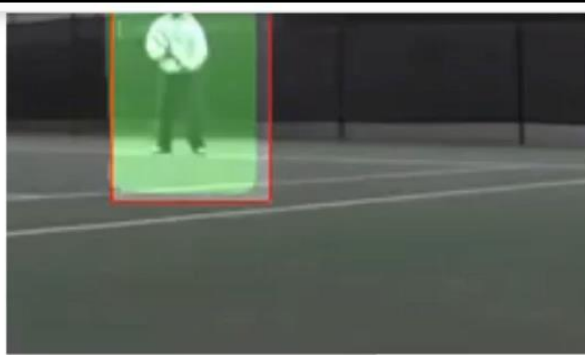


Shevtsov, Alexander, et al. "Analysis of Twitter and YouTube during US elections 2020."
arXiv e-prints (2020): arXiv-2010.

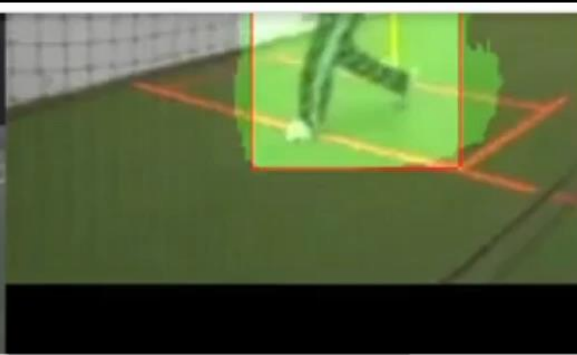
Semi-supervised learning

- The model is initially trained with a **small amount of labeled data** and a **large amount of unlabeled data**.





TennisSwing: 0.851



RopeClimbing: 0.855



Biking: 0.84

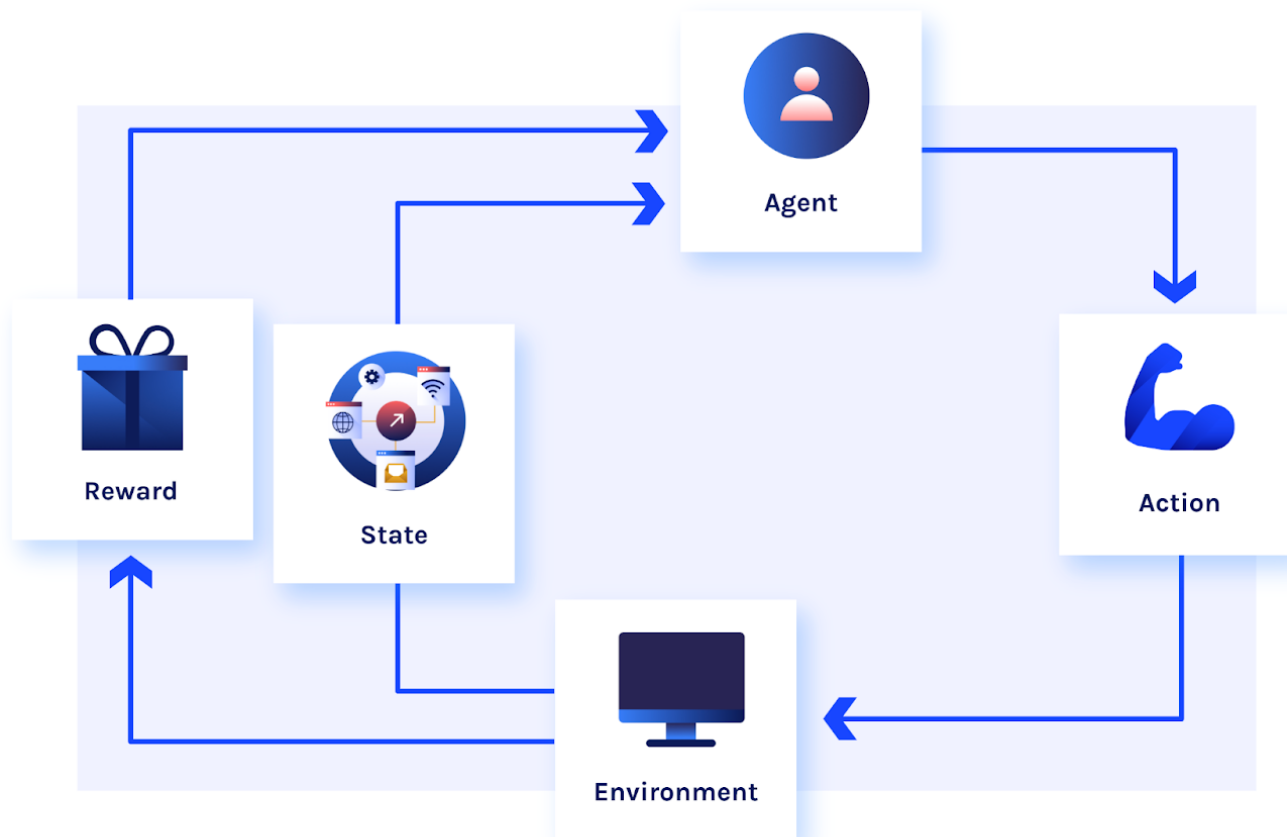


Human action recognition problem

Semi-Supervised Learning
and Capsule Network for
Video Action Detection.
RIVF 2023. (HCMUS)

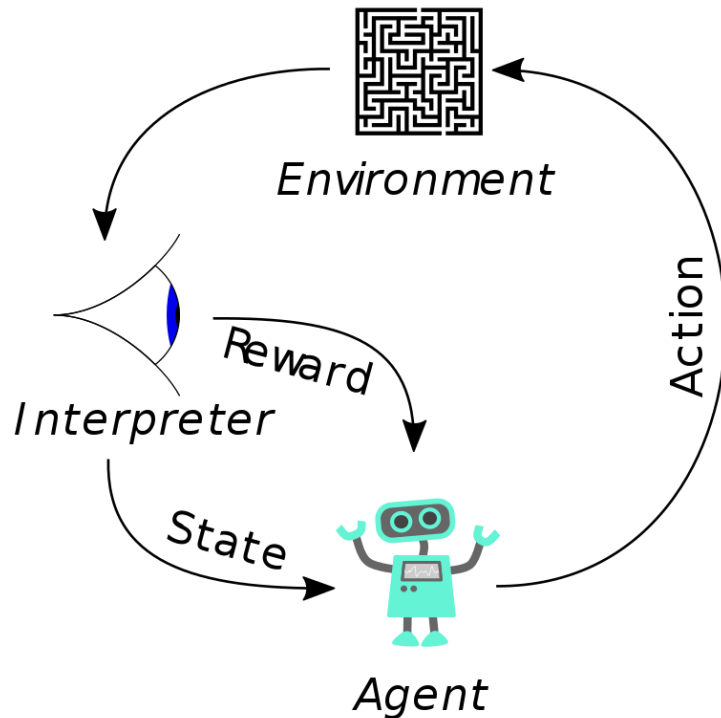
Reinforcement learning (RL)

- The agent learns from the environment by **interacting** with it and **receives rewards** for performing actions.



Reinforcement learning

- The agent learns from the environment by interacting with it and receives rewards for performing actions.



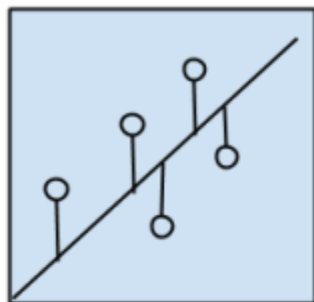
Learning to ride a bike requires trial and error, much like reinforcement learning. (Video courtesy of Mark Harris, who says he is “learning reinforcement” as a parent.) 18

Reinforcement learning: Demo

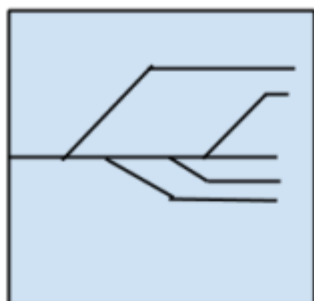


Source: [YouTube video](#)

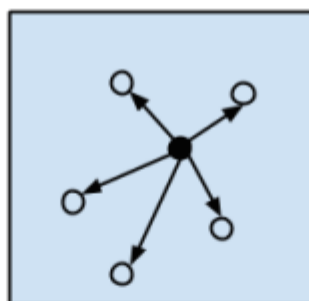
Machine learning algorithms



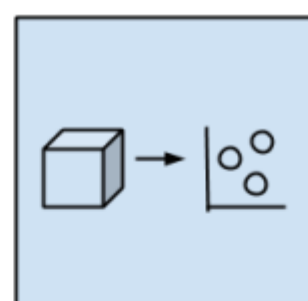
Regression Algorithms



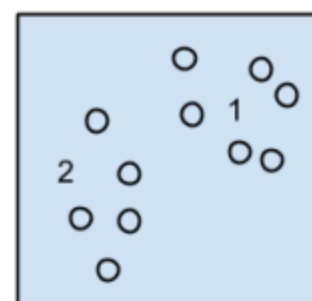
Regularization Algorithms



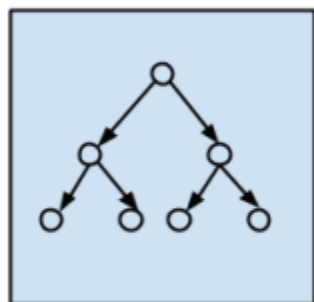
Instance-based Algorithms



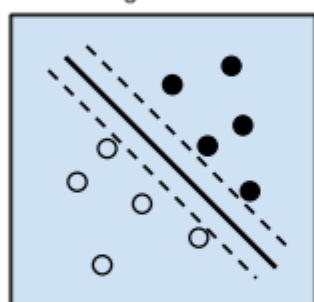
Dimensional Reduction Algorithms



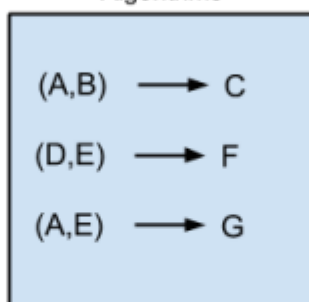
Clustering Algorithms



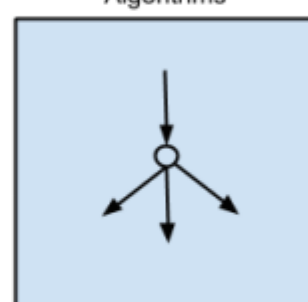
Decision Tree Algorithms



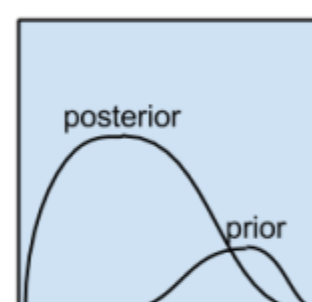
Support Vector Machines



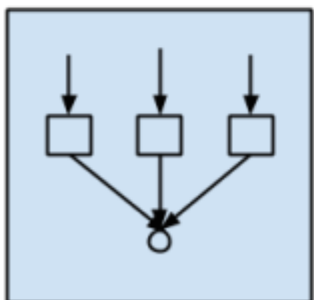
Association Rule Learning Algorithms



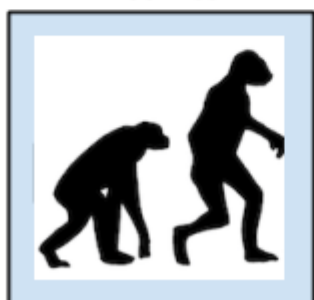
Artificial Neural Network Algorithms



Bayesian Algorithms

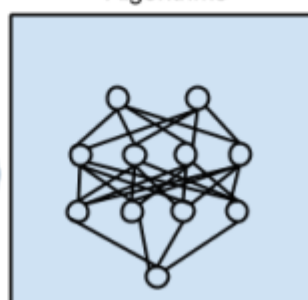


Ensemble Algorithms

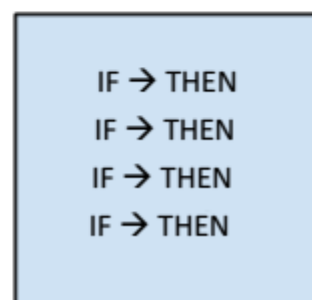


Evolutionary Algorithms

Non-exhaustive
list of ML families



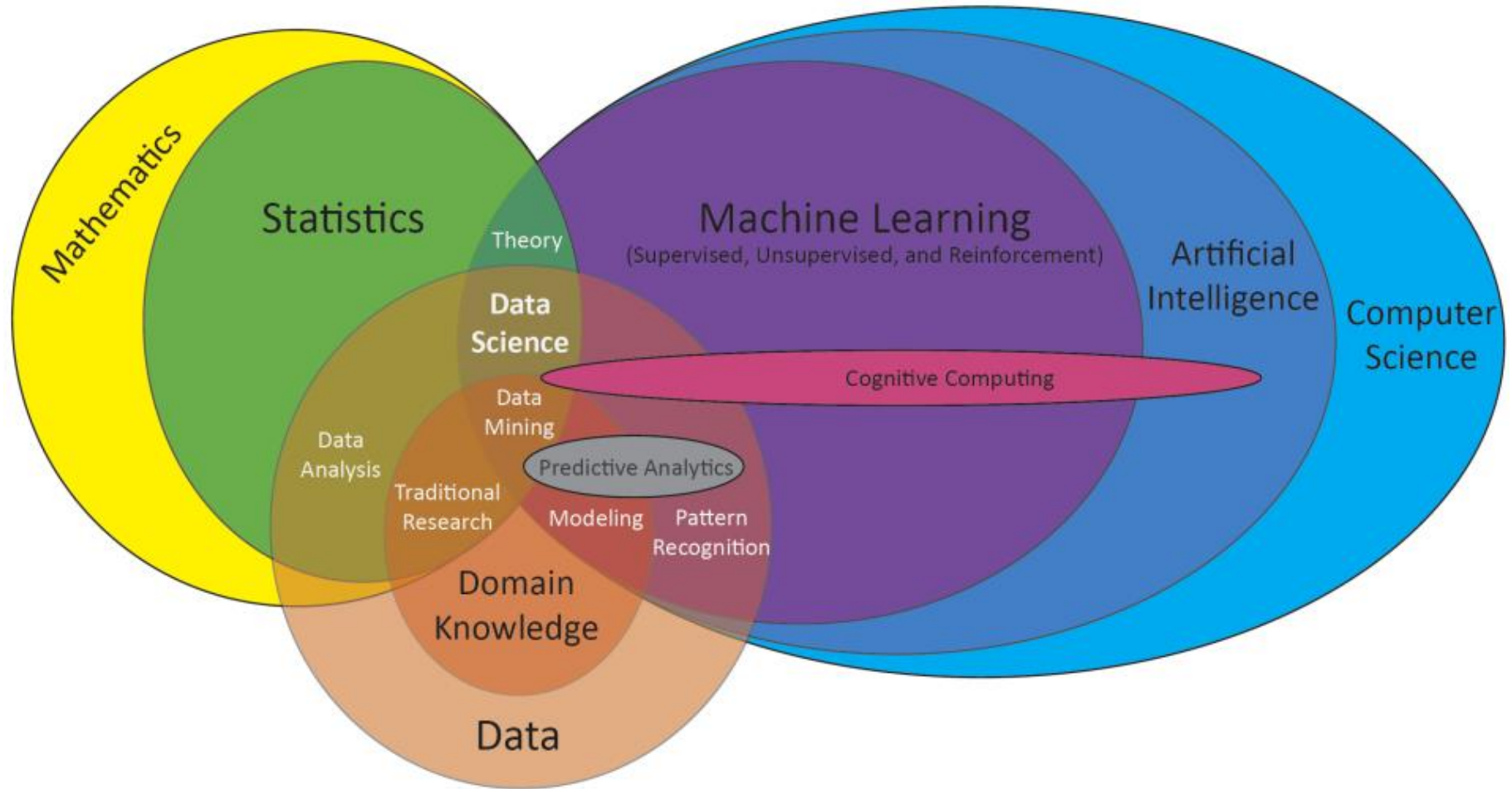
Deep Learning Algorithms



Learning Classifier Systems

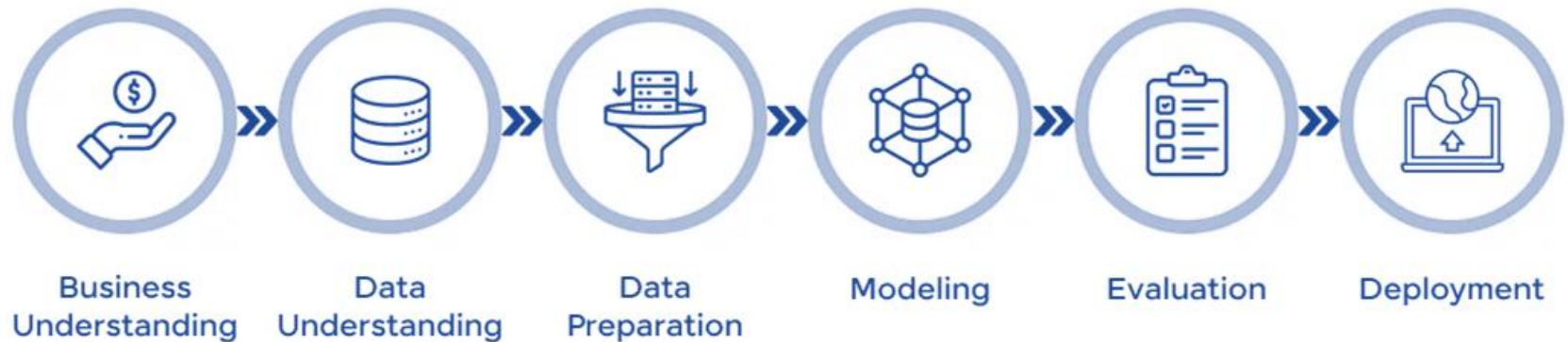
Source: [An introduction to Machine learning](#)

Machine learning and related areas



Machine learning and related areas

- Machine learning serves a key step in Data science process, providing tools to analyze data for better insights.



- It also motivates the studies of new learning algorithms to meet the challenges of massive data.

Acknowledgements

- Some parts of the slide are adapted from
 - Maria-Florina Balcan, Introduction to Machine Learning, 10-401, Spring 2018, Carnegie Mellon University
 - Ryan Urbanowicz, An Introduction to Machine Learning, PA CURE Machine Learning Workshop: December 17, School of Medicine, University of Pennsylvania



...the end.

